



Dental Cements (Part -1)



Learning outcomes

- Classify Dental Cements
- Explain zinc oxide eugenol cement composition, setting reaction, properties and uses.
- Explain zinc phosphate cement, composition, setting reaction, properties and uses.

Introduction

- A dental cement is a substance that hardens to act as a Base, Liner, Filling material or Adhesive to Bind Devices and Prosthesis To Tooth Structure To Each Other .

(Anusavice)

General uses of dental cements:

Liners & Bases

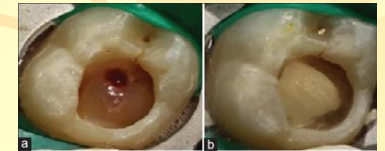
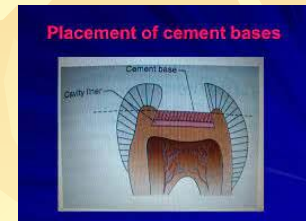
Restoration

Pulp capping

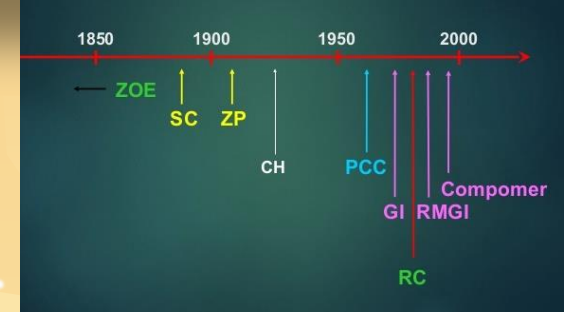
Obtundant

Luting /cementation of prosthesis and orthodontic appliance

Periodontal pack, Root canal sealers



History...



- 1873- Silicate cement by Fletcher
- 1890- Zinc oxide and clove oil by Chisolm
- 1879- Zinc phosphate cement by Dr Pierce
- 1930- Calcium hydroxide paste by Hermann
- 1968-Polycarboxylate cement by Dennis Smith
- 1971-Glass ionomer cement by Wilson and Kent
- 1992 – Resin modified GIC
- 2004 – Resin cement



Classification of Dental Cements

According to Craig

- Water based cements
 - Glass ionomer cements
 - Zinc Polycarboxylate
 - Zinc Phosphate

- Resin based cements
 - Adhesive Resin cements
 - Compomers

- Oil based cements
 - Zinc Oxide Eugenol
 - Non Eugenol -Zinc Oxide

Ideal requirements of dental cements

- Able to protect pulp
- Should be insoluble in saliva & liquids
- Should be dimensionally stable.
- Should be Non porous.
- Should be biocompatible and non irritant.
- Should be easy to manipulate.



ZINC OXIDE EUGENOL CEMENT

- Zinc oxide eugenol (ZOE) cement have been used extensively in dentistry since 1890s.
- They are cements of Low strength.
- Also they are the Least irritating of all dental cements and are known to have an Obtundant(Reduce dentin sensitivity)effect on exposed dentin.

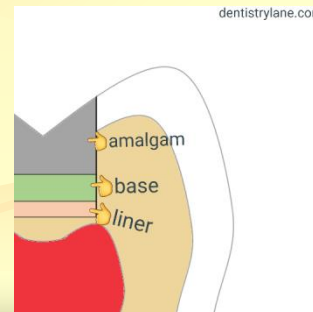


CLASSIFICATION

American D Association

ADA specification no.30 listed 4 types

- Type I ZOE: For temporary Cementation
- Type II ZOE: Permanent Cementation
- Type III ZOE: Temporary filling and thermal base
- Type IV ZOE: Cavity Liners



Mode of supply

Dispensed in two forms:

- Powder and liquid



- Two Paste system



Composition:

Powder:

- ZnO- 69.0%; Principal ingredient
- White rosin- 29.3%; to reduce brittleness of set cement.
- Zn Stearate- 1.0%; Accelerator
- Zn Acetate- 0.7%; Improves strength.

Liquid:

- Eugenol- 85.0%; reacts with ZnO.
- Olive oil- 15%; plasticizer

Setting reaction:

- In first reaction hydrolysis of ZnO to its hydroxide takes place.



- Then it proceeds as acid-base reaction to form chelate.



Base

Acid

Salt

Set cement consists of particles of ZnO embedded in matrix of Zn eugenolate.

Properties:

1. Mechanical properties:

A) Compressive strength:

- Weak cement
- Compressive Strength ranges from 3 Mpa to upto 55 Mpa.

B) Tensile Strength:

- Ranges from 0.32 to 5.8 Mpa.

2) *Film thickness:* 25 micron

Important for those cements used for cementation of restoration.

3) *Biologic properties:*

A) pH and effect on pulp:

- Least irritating.
- pH: 6.6-8.0

B) Bacteriostatic and Obtundant

4) *Adhesion:*

- Do not adhere to enamel or dentin

5) *Optical properties:*

- Opaque

Manipulation:

- ***P/L ratio:*** 4:1 by wt.
- Bottles are shaken gently.
- Measured quantity of powder and liquid dispensed onto cool glass slab.
- Bulk of powder incorporated in liquid and spatulated thoroughly in circular motion with stainless steel spatula.
- Small increments then added till mix is complete.

Setting time: 4 -10min.



FACTORS AFFECTING SETTING TIME

- Particle size: Smaller zinc oxide particles set faster
- Powder to liquid ratio: Higher the ratio, faster the set
- Cooling the glass slab: Slows the setting reaction and delays the setting time

Modified ZnO Eugenol Cements:

1. EBA Alumina Modified Cements:

- Introduced to Improve mechanical properties.

Composition:

Powder:

- ZnO- 70%
- Alumina- 30%

Liquid:

- EBA(Ethoxy Benzoic acid)- 62.5%
- Eugenol- 37.5%



2. Polymer Reinforced ZnO Eugenol

- Introduced to improve mechanical properties.

Composition:

Powder:

- ZnO
- Finely divided natural or synthetic resins

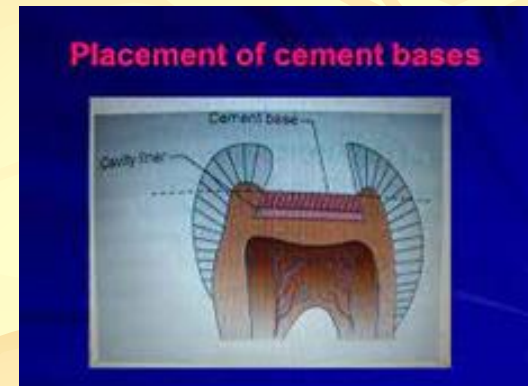
Liquid:

- Eugenol
- Acetic acid- Accelerator
- Thymol – Antimicrobial agent



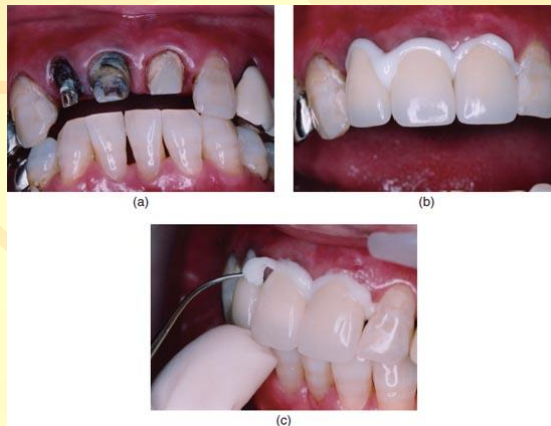
Uses of zinc oxide eugenol

- For Temporary cementation
- For Temporary fillings.
- Cavity Liner and Base
- Pulp capping agent



Zinc Phosphate Cement

- Oldest of luting cements.
- ‘Crown and bridge’ cement and ‘Zinc Oxyphosphate’ have also been used.



Classification:

A.D.A specification no. 8 designate them as:

- **Type I:** Fine grained for Luting

Film thickness- 25 micron or less

- **Type II:** Medium grain for Luting and Filling

Film thickness- 40 micron

Mode of supply:

available as

- Powder and liquid system
- Capsules of pre-proportioned powder liquid.



Composition:

Powder :

- Zinc oxide -92%
- Magnesium oxide - 8.2%

Liquid:

- Phosphoric acid- 38.2%- reacts with ZnO
- Water - 36% -controls rate of reaction

Setting Reaction:

Powder + liquid $\longrightarrow$ Zinc Aluminophosphate

- Set cement is a cored structure composed of unreacted zinc oxide particles embedded in amorphous matrix of zinc Aluminophosphate

Setting time: 5-9 min

Properties:

1. Mechanical property

Compressive strength:

- Stronger than Zinc oxide eugenol Cements
- Max strength attained in 1st day.(103.5 Mpa)
- It is sufficient, when used as base or luting agent.

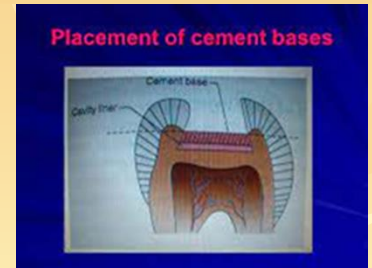
2Tensile strength:

- Weaker in tension, Brittle (5.5 Mpa)

2. Film thickness:

Type 1 - 25 micron

Type 2 - 40 micron



3. Biologic properties:

- Acidity high due to phosphoric acid, 3 min after mixing pH is 3.5, Reaches neutral in 24 - 48 hours..
- Pulpal irritation if dentin thickness is less
- To prevent pulpal irritation, calcium hydroxide liner can be used

4. Adhesion property:

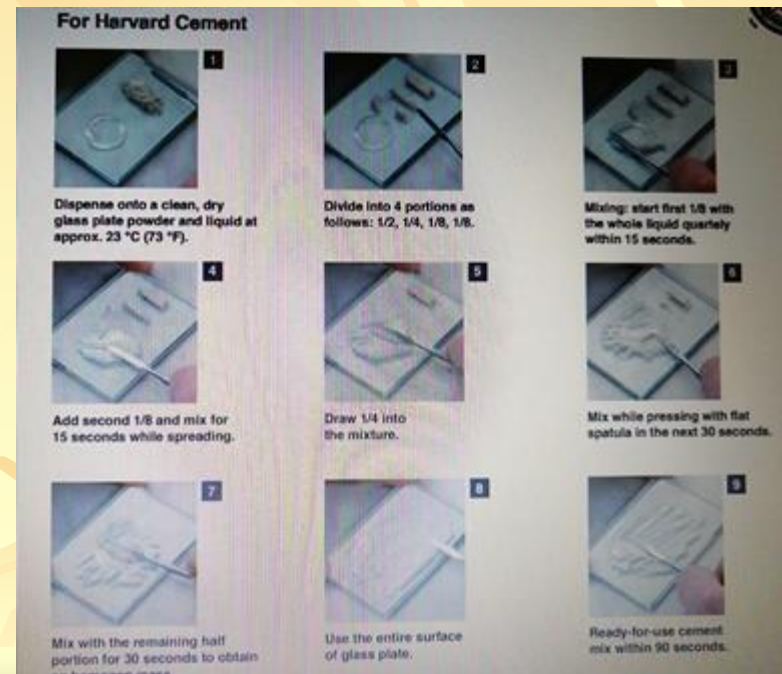
- No chemical bond, depends on mechanical interlocking of set cement.

5. Optical property:

- Set cement- opaque

Manipulation:

- Cool glass slab to delay setting and allow more powder to be incorporated.
- Powder added in small increments
- Large area covered during mixing in order to dissipate exothermic heat.



Manipulation Procedure:

- ***P/L ratio:*** 1.4gm /0.5ml.
- Powder divided into small increments.
- Stainless steel spatula used.
- Mixing initiated by addition of small amount of powder at a time in liquid.
- Spatulation carried out with brisk, circular motion of spatula, covering large area.

Mixing time: 1min 15 sec. ***Setting time:*** 5-9 min



Learning Resources

- Craig & Powers, “Restorative Dental Materials” 11th ed. (Mosby, St. Louis).
Pages: 595 - 611



Thank you